Decodability Is Not Validity: Evaluating Nuisance Sensitivity in Language Model Concept Probes

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https://github.com/kleinay/decodability-is-not-validity

Abstract

Two concept probes can be equally accurate at predicting a concept from a language model's hidden states while their scores contain different mixtures of the target concept and a correlated nuisance attribute. We measure how often this happens and what it costs. Across three public datasets that pair a target with a different kind of nuisance (two aspect sentiments; occupation and gender; toxicity and identity mention), two LLMs, and a population of concept extractors, we compare decodability with discriminant evidence about one named nuisance: the score's dependence on the nuisance with the target held fixed, and its response to input edits intended to change only the nuisance, filtered by two LLM judges and compared against placebo edits. Under correlated training data, extractors that are statistically equivalent in held-out AUROC differ several-fold in nuisance sensitivity; decodability is not a reliable proxy for it, and its relationship to nuisance sensitivity changes sign across settings. Where target edits exist (CEBaB), competitive extractors pass them regardless of nuisance sensitivity. Selecting extractors with nuisance-aware development diagnostics, which need nuisance labels but no counterfactuals, lowers counterfactual nuisance sensitivity at a modest accuracy cost. Concept measurements should be validated, not just decoded.

1 Introduction

Probes, concept activation vectors and subspace readouts turn a hidden state into a scalar proxy for a concept T in the input (Alain and Bengio, 2017; Kim et al., 2018; Belinkov, 2022; Zou et al., 2023; Marks and Tegmark, 2024); the usual evidence for that reading is decodability, i.e. predicting held-out labels of T. Decodability cannot identify *what* the score measures. For binary T and a nuisance N, the scores $s_1 = T$ and $s_2 = T + \alpha N$ with $0 < \alpha < 1$ both rank T perfectly, yet only s_2 moves when N alone changes. When two extractors are equally accurate, nothing in their accuracy says which one is more selective for T.

Measurement theory treats this as a question of construct validity, an evidential and multidimensional property of a score's interpretation rather than a number the score has (Cronbach and Meehl, 1955; Messick, 1995; Jacobs and Wallach, 2021; Bean et al., 2025). Part of that evidence is *discriminant*: the score should reflect the intended construct rather than a distinct, possibly correlated one (Campbell and Fiske, 1959). Valid measures of correlated constructs can correlate; what discriminant evidence asks is whether the score depends on the other construct *once the target is held fixed*. We study exactly this, for one named nuisance per dataset; passing our tests is necessary evidence about that nuisance, not evidence of validity in general.

That accuracy cannot detect such dependence is not new: probes trained on spuriously correlated data can rely on the spurious attribute (Kumar et al., 2022), probing accuracy need not reflect use by the model (Hewitt and Liang, 2019; Ravichander et al., 2021; Elazar et al., 2021), and interventions on representations trade completeness against selectivity (Canby et al., 2025; Huang et al., 2024). Recent work tests whether probes and SAEs disentangle known concepts (Mueller et al., 2026), compares concept-subspace estimators (Naowarat et al., 2026), shows accurate probes can be misaligned (Lysnæs-Larsen et al., 2025), and links embedding-based social measures to confounding and counterfactual validation (Li et al., 2026). We ask a narrower question about scalar readouts: among extractors that are accuracy-matched on held-out data, how much does sensitivity to a named nuisance under input edits vary, does it persist without training-set correlation, and can development diagnostics choose better? We answer with a population study over

three variable types and two models, using judge-filtered, placebo-controlled input edits (§4), and a comparison of development-set selection criteria (§4.1). Our interventions edit the *input text* of frozen models; we make no claim that the extracted directions causally mediate model behavior.

2 Measuring nuisance selectivity

An *extractor* maps a hidden state h to $s(h) = w^\top z(h) + b$, with z the standardization on training data; subspace methods have a linear readout too, so every extractor is a scalar score. Each dataset supplies a binary target T, a binary nuisance N correlated with T in natural data, and edits of the input. Let $\rho = P(N{=}1 \mid T{=}1) = P(N{=}0 \mid T{=}0)$ with balanced T. Extractors are trained at $\rho_{\text{tr}} \in \{0.5, 0.7, 0.9\}$; test metrics reweight the four (T, N) cells of one fixed test pool.

DEC: AUROC of s for T at $\rho = \rho_{\text{tr}}$ (in-distribution decodability, the standard criterion).

$|$**LEAK**$|$: $\frac{1}{2}\sum_t |\text{AUROC}(s \to N \mid T{=}t) - 0.5|$, dependence on N with the target held fixed. The absolute form avoids cancelling opposite dependences in the two target classes.

CONFLICT: AUROC for T on the cells with $T \neq N$ (a fully reversed correlation). A stress test, not a validity measure: there, $s = -N$ also ranks T perfectly.

CF$_N$: mean change in s under edits that flip only N, signed toward the N value that co-occurs with $T{=}1$, divided by the mean score gap between the T classes on the decorrelated pool (Δ_T). As a signed mean can hide opposite changes, we also report $|\text{CF}_N|$ (mean absolute change over Δ_T) and compare with *placebo* edits that change neither variable.

CF$_T$: the same quantity for edits that flip T (CEBaB only), i.e. convergent counterfactual evidence.

3 Setup

Data and edits (Table 1). **CEBaB** (Abraham et al., 2022): restaurant reviews with aspect labels and human-written counterfactual edits; we pair each original test review with edits that flip food (or, for CF$_T$, service) sentiment and leave the other aspect's label unchanged; splits are disjoint by review family, and bootstrap CIs resample families. **BiasBios** (De-Arteaga et al., 2019): surgeon vs. nurse biographies; N is the subject's gender, edited by swapping pronouns and gendered nouns in both directions. **CivilComments** (WILDS; Borkan et al., 2019; Koh et al., 2021): toxicity, with N a human annotation that the comment mentions Muslims or LGBTQ people; we swap those identity terms for Christian/straight terms and, on 400 further comments, "Christian" for "Muslim" (Dixon et al., 2018; Garg et al., 2019), and average the two directions. We keep up to 1,100/250/400 training/development/test texts per (T, N) cell and train on samples of 2,000. Correlated regimes are realistic: in BiasBios, $P(\text{male} \mid \text{surgeon}) = 0.85$ and $P(\text{female} \mid \text{nurse}) = 0.91$; in unedited CEBaB reviews, 0.80 and 0.73.

	target T / nuisance N	N edits	judged by both	accepted
CEBaB	service / food sentiment	210	210	41%
BiasBios	surgeon vs. nurse / gender	1589	800	72%
Civil	toxic / Muslim or LGBTQ	762+400	762+400	13%, 29%

Table 1: Datasets: the variable pairs differ in kind (two evaluative aspects of one text; an occupation and an attribute of its subject; a pragmatic property and a topic). Edits: generated pairs; judged by both: pairs rated by both LLM judges (BiasBios: a random subset); accepted: share of jointly judged pairs both accept (§3). Civil: forward/reverse identity swaps.

Edit validation. Two distinct judge models, neither identical to a probed model (Qwen3-32B, run locally, which shares a model family with Qwen2.5-7B; and gpt-oss-120b), see each pair in random order, without being told which text is edited or which attribute should change, and say whether the target and the nuisance are the same, rate each text's fluency, and flag other meaning changes. Acceptance is a conservative filter, not verified correctness: a nuisance edit is *accepted* if both judges find the target unchanged and the nuisance changed, with no major other meaning change and no fluency drop. All counterfactual results use accepted pairs only (Appendix A). Identity swaps are often rejected for changing a claim about a group, not just its name (Table 1). Placebo edits replace neutral words with synonyms (all datasets), or "Christian" with "Protestant" (Civil).

Models and extractors. Qwen2.5-7B-Instruct (Qwen Team, 2024) and Gemma-2-9B-it (Gemma Team, 2024), frozen; residual stream mean-pooled over non-special tokens at 7 depths. Per layer: difference of class means; ℓ_2 logistic regression ($C \in \{10^{-3}, \ldots, 10\}$); partial least squares ($k \in$

		best DEC	#near (equiv.)	CF_N near-best range	CF_N equiv. range	CONFLICT range
$\rho_{tr} = 0.9$ *(correlated training data)*						
CEBaB	Q	.966	37 (9)	.46–.89	.54–.81	.44–.69
	G	.979	39 (8)	.26–.72	.25–.63	.57–.82
BiasBios	Q	.998	66 (66)	.08–.41	.08–.41	.83–.97
	G	.998	66 (63)	.08–.29	.09–.29	.92–.97
Civil	Q	.932	8 (4)	.24–.45	.28–.37	.35–.41
	G	.940	12 (1)	.30–.50	.35	.29–.43
$\rho_{tr} = 0.5$ *(decorrelated)*						
CEBaB	Q	.924	10 (1)	.02–.28	.09	.84–.90
	G	.960	11 (2)	−.10–.08	.01–.02	.93–.95
BiasBios	Q	.995	58 (54)	−.04–.04	−.03–.05	.98–.99
	G	.995	60 (55)	−.07–.03	−.06–.03	.98–.99
Civil	Q	.829	4 (2)	.04–.08	−.00–.03	.81–.83
	G	.839	5 (2)	−.02–.07	−.03–−.02	.83–.84

Table 2: Standard extractors that are near-best (test DEC within 0.01 of the best) and, in parentheses, statistically *equivalent* to the best (TOST, margin 0.01). Q: Qwen2.5-7B; G: Gemma-2-9B. $\rho_{\text{tr}} = 0.7$: Appendix B.

$\{2, 4, 8, 16\}$; k=1 equals the mean difference and is dropped); and a *nuisance-erased* probe that projects out the within-class nuisance mean differences, estimated on the training sample only (cf. Ravfogel et al., 2020; Belrose et al., 2023) before logistic regression. The first three families use no nuisance labels (*standard*, 70 per setting); the erased probe adds 35. Each is fit on three training samples. An extractor is *near-best* if its test DEC is within 0.01 of the best, and *equivalent* to the best if a paired-bootstrap 90% CI of the DEC difference lies inside ± 0.01 (TOST at $\alpha = 0.05$). Because the reference is selected on the test pool, these equivalence results are descriptive; the selection comparisons (§4.1) choose on development data only.

4 Results

Accuracy-matched extractors differ in nuisance sensitivity. With correlated training data ($\rho_{\text{tr}} = 0.9$), extractors equivalent to the best in decodability span CF_N from .54 to .81 on CEBaB (Qwen; Gemma .25–.63): for some, rewriting only the food sentence of a review shifts the "service" score most of the way to the other service class. On BiasBios DEC is at ceiling for nearly all extractors (66 equivalent to the best), while pronoun swaps shift the occupation score by .08–.41 of the surgeon–nurse gap. On Civil few extractors pass the strict equivalence test, but every near-best one has CONFLICT < 0.5: the most decodable toxicity scores rank non-toxic comments about Muslims or LGBTQ people above toxic comments that mention neither. All CF_N values use accepted edits; extractor rankings agree with those on all edits ($\geq$.77) and under SD normalization ($\geq$.87), and signed and absolute CF_N agree at $\rho_{\text{tr}} \geq 0.7$ ($\geq$.89; Appendix A). Placebo edits barely move scores (median $|CF_N|$ .01–.07, vs. .15–.60 for nuisance edits at $\rho_{\text{tr}} = 0.9$).

With decorrelated training, the directional effect mostly disappears. At $\rho_{\text{tr}} = 0.5$, near-best extractors show little net nuisance shift on BiasBios and Civil (signed CF_N within ± 0.08 on accepted edits) and less on CEBaB (.02–.28, Qwen). Example-level shifts remain on CEBaB, though: median $|CF_N|$ is .25 against a signed .07 (Qwen; placebo .06, for smaller edits). Training-set correlation drives the systematic reliance on the nuisance. A second source is geometric: on CEBaB the target direction and the within-class food direction are highly aligned (cosine .66–.83 across layers and models; BiasBios .06–.18), and this observational geometry alone approximately predicts the counterfactual food sensitivity of the difference-of-means direction trained on decorrelated data (predicted .76–.92, observed .63–.85; mean absolute error .12; BiasBios .02). Discriminative probes are much less affected. In a synthetic representation, overlap raises CF_N far more than it changes decodability (Appendix C).

Decodability is not a reliable proxy, and on CEBaB target edits do not arbitrate. Among competitive standard extractors at $\rho_{\text{tr}} = 0.9$, the rank correlation between DEC and CF_N ranges from −.61 to +.39 and changes sign across settings (descriptive bootstrap intervals in Appendix B). On CEBaB every near-best extractor, in every regime, responds fully to genuine *service* edits (CF_T .84–1.19) while its food sensitivity ranges over −.10–.89. Here only nuisance edits separate more from less selective scores.

A fixed-method comparison. Choosing each family's layer and hyperparameter on development data, the logistic probe and the nuisance-erased probe are equivalent in DEC on BiasBios at $\rho_{\text{tr}} = 0.9$ (difference +.001, 90% CI [+.000, +.002], Qwen) yet CF_N is .20 vs. .06 (Gemma: .20 vs. .06). On CEBaB ($\rho_{\text{tr}} = 0.9$) equivalence is not established (CI up to +.023) while CF_N falls from .83 to .34 (Appendix B). Erasure lowers but does not remove nuisance sensitivity: linear guarding on training data is not invariance to edits.

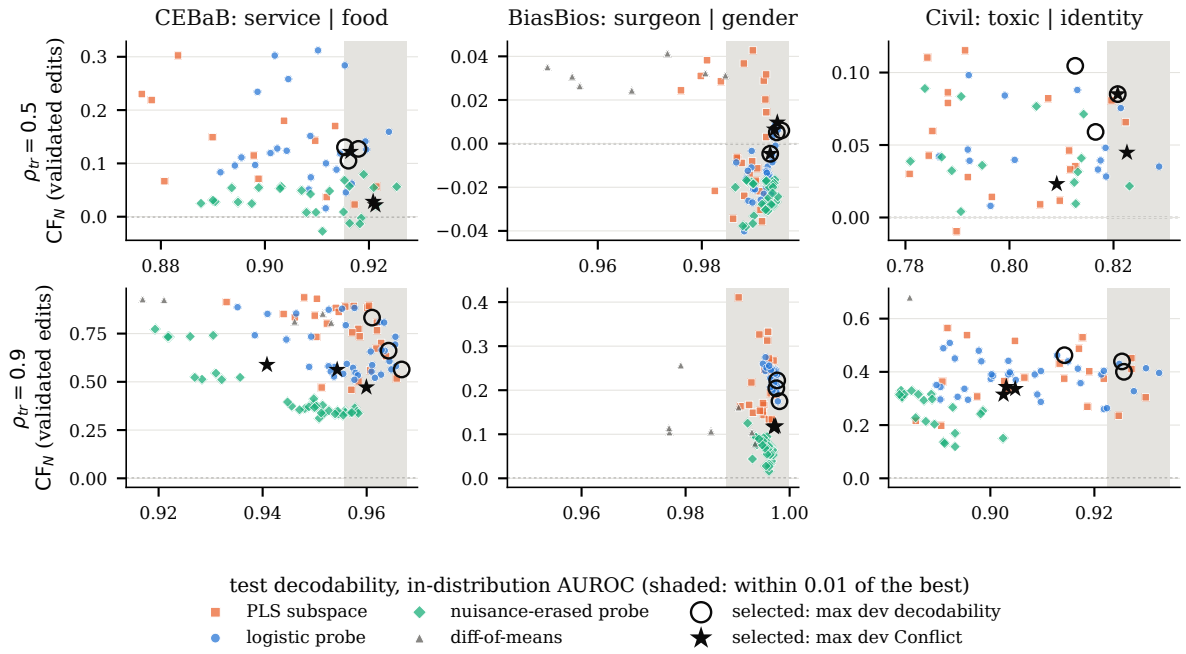


Figure 1: Competitive extractors (DEC within 0.05 of the best; Qwen2.5-7B; mean over 3 training samples): test decodability vs. nuisance sensitivity CF$_N$ on accepted edits (0: no net shift; 1: the nuisance edit shifts the score as much as a change of target class). Black symbols: the extractor that each training sample selects on development data by maximal DEC (ring) or maximal CONFLICT (star), standard extractors only. Gemma-2-9B: Appendix B.

		standard extractors			+ erased
		max DEC	max CONFLICT	min |LEAK|	max CONFLICT
CEBaB	Q	.69	.54 (−.012)	.50 (−.013)	.34 (−.007)
	G	.41	.26 (−.005)	.27 (−.004)	.11 (−.004)
BiasBios	Q	.20	.12 (−.001)	.08 (−.004)	.03 (−.002)
	G	.22	.12 (−.002)	.11 (−.006)	.02 (−.002)
Civil	Q	.43	.33 (−.018)	.31 (−.001)	.15 (−.034)
	G	.31	.20 (−.015)	.30† (−.006)	.11 (−.036)

Table 3: Selecting an extractor on development data at $\rho_{\text{tr}} = 0.9$: test CF$_N$ (accepted edits) of the extractor chosen by maximal DEC (standard practice), maximal CONFLICT, or minimal |LEAK| among extractors within 0.01 of the best development DEC; change in test DEC in parentheses. Means over 3 training samples; †: CF$_N$ did not fall in every sample. More criteria: Appendix B.

4.1 Selecting for nuisance selectivity

Practitioners choose a layer and hyperparameters by development accuracy. Table 3 compares this with nuisance-aware criteria that need nuisance labels on development data but no counterfactuals. Among standard extractors, choosing by CONFLICT lowers CF$_N$ in every dataset, model and training sample (by .08–.15) at a mean DEC cost of at most .018 (.023 in the worst training sample); it lowers |CF$_N$| as well, so it is not selecting scores that rely on the opposite of the nuisance. The leakage-constrained rule is similarly effective except on Civil with Gemma; a combined DEC−|LEAK| rule behaves similarly (Appendix B). When the nuisance-erased probe is available, CONFLICT and DEC−|LEAK| select it in every sample (the leakage-constrained rule in 67%), and with CONFLICT CF$_N$ falls by .17–.34 at a mean DEC cost of at most .036 (on Civil, where the nuisance is predictive in-distribution). These accuracy costs are modest, not zero.

5 Discussion

Our claim is narrower than "probes are invalid": in-distribution decodability is not a reliable proxy for selectivity with respect to a named nuisance. Equally decodable scores can carry very different amounts of nuisance, mainly through correlations in the probe's training data and, in CEBaB, overlapping target and nuisance geometry. A claim that a direction *measures* a concept should name plausible nuisances and report discriminant diagnostics and, where possible, nuisance-only edits; on CEBaB, target edits were passed regardless. Low nuisance sensitivity is necessary discriminant evidence about one nuisance, not construct validity (Messick, 1995; Licht et al., 2025).

Limitations

Each dataset has one nuisance; selectivity for unexamined nuisances is unknown. Edits are filtered by two LLM judges rather than validated by humans; the judges agree poorly to moderately, and accepted pairs may be a selective subset (Appendix A); the code generates a blinded stratified sample for human annotation. Rule-based edits leave other cues (first names; the topic of religion or sexuality), and accepting only some edits may select easier pairs, although results on all and accepted pairs agree. CF_N is normalized by the class score gap, which is unstable for weak extractors, so we analyze competitive ones; an effect-size normalization (by the within-cell score SD, akin to Cohen's d) ranks extractors similarly ($\geq$.87). Correlated regimes are produced by resampling; they match natural rates for BiasBios and approximately CEBaB, and are a stress test for Civil. Strict equivalence is established for few extractors on CEBaB and Civil. We study linear readouts of mean-pooled (and, as an ablation, last-token) representations of two 7–9B models; nonlinear probes, SAE features and other models are untested. Our measures concern what a score tracks in text, not what the model computes with it.

Ethics Statement

Two datasets involve gender and religious or sexual identity. Concept scores used for auditing or monitoring can encode such attributes while passing accuracy checks; our diagnostics complement, and do not replace, fairness evaluation. The gender swap follows the dataset's binary annotation.

References

Eldar David Abraham, Karel D'Oosterlinck, Amir Feder, Yair Ori Gat, Atticus Geiger, Christopher Potts, Roi Reichart, and Zhengxuan Wu. 2022. CEBaB: Estimating the causal effects of real-world concepts on NLP model behavior. In *Advances in Neural Information Processing Systems*.

Guillaume Alain and Yoshua Bengio. 2017. Understanding intermediate layers using linear classifier probes. In *ICLR Workshop*.

Andrew M. Bean, Ryan Othniel Kearns, Angelika Romanou, Franziska Sofia Hafner, Harry Mayne, et al. 2025. Measuring what matters: Construct validity in large language model benchmarks. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*.

Yonatan Belinkov. 2022. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. 2023. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems*.

Daniel Borkan, Lucas Dixon, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2019. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion Proceedings of The 2019 World Wide Web Conference*, pages 491–500.

Donald T. Campbell and Donald W. Fiske. 1959. Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin*, 56(2):81–105.

Marc E. Canby, Adam Davies, Chirag Rastogi, and Julia Hockenmaier. 2025. How reliable are causal probing interventions? In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 857–878.

Lee J. Cronbach and Paul E. Meehl. 1955. Construct validity in psychological tests. *Psychological Bulletin*, 52(4):281–302.

Maria De-Arteaga, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. 2019. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 120–128.

Lucas Dixon, John Li, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2018. Measuring and mitigating unintended bias in text classification. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, pages 67–73.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. 2021. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175.

Sahaj Garg, Vincent Perot, Nicole Limtiaco, Ankur Taly, Ed H. Chi, and Alex Beutel. 2019. Counterfactual fairness in text classification through robustness. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*, pages 219–226.

Gemma Team. 2024. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*.

John Hewitt and Percy Liang. 2019. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP*, pages 2733–2743.

Jing Huang, Zhengxuan Wu, Christopher Potts, Mor Geva, and Atticus Geiger. 2024. RAVEL: Evaluating interpretability methods on disentangling language model representations. In *Proceedings of ACL*.

Abigail Z. Jacobs and Hanna Wallach. 2021. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 375–385.

Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. 2018. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In *Proceedings of ICML*, pages 2668–2677.

Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao, et al. 2021. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of ICML*, pages 5637–5664.

Abhinav Kumar, Chenhao Tan, and Amit Sharma. 2022. Probing classifiers are unreliable for concept removal and detection. In *Advances in Neural Information Processing Systems*.

Baishi Li, Ta Yu, Kelvin J. L. Koa, and Ke-Wei Huang. 2026. The proxy presumption: From semantic embeddings to valid social measures. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 22892–22910.

Hauke Licht, Rupak Sarkar, Patrick Y. Wu, Pranav Goel, Niklas Stoehr, Elliott Ash, and Alexander Miserlis Hoyle. 2025. Measuring scalar constructs in social science with LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 32144–32171.

Jacob Lysnæs-Larsen, Marte Eggen, and Inga Strümke. 2025. Probing the probes: Methods and metrics for concept alignment. *arXiv preprint arXiv:2511.04312*.

Samuel Marks and Max Tegmark. 2024. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Conference on Language Modeling (COLM)*.

Samuel Messick. 1995. Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50(9):741–749.

Aaron Mueller, Andrew Lee, Shruti Joshi, Ekdeep Singh Lubana, Dhanya Sridhar, and Patrik Reizinger. 2026. From isolation to entanglement: When do interpretability methods identify and disentangle known concepts? In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 17188–17210.

Burin Naowarat, Hao Tang, and Sharon Goldwater. 2026. A framework for analyzing concept representations in neural models. In *Proceedings of the 30th Conference on Computational Natural Language Learning*, pages 574–587.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. 2020. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of ACL*, pages 7237–7256.

Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. 2021. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of EACL*, pages 3363–3377.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. 2023. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*.

A Edit validation and robustness

Table 4 reports, per judge and edit kind, how often the judge found the target kept, the nuisance changed, a fluency drop (edited text rated non-fluent while the source was rated fluent; both judges rate many untouched informal texts as non-fluent, so absolute fluency is not used), and a major other meaning change. Table 5 gives the agreement of the two judges' accept decisions: agreement is low to moderate (κ .05–.42), mostly because Qwen3-32B is much stricter (it accepts .44–.60 of placebo edits, gpt-oss-120b .92–.98), often reporting a fluency drop or a changed target for synonym swaps. Requiring both judges is therefore conservative, and acceptance by both is a filter, not verified correctness; rankings on accepted and on all edits agree closely (Table 6). For Civil, Qwen3-32B judges that swapping the named group changes toxicity in about half of the forward swaps, so few identity swaps survive. The judges were never told which text was edited or which attribute was meant to change. The released code generates a blinded stratified sample of 25 pairs per dataset and edit kind for human annotation.

data	edit kind	n	judge	T kept	N changed	fluency drop	major other	accepted
BiasBios	nuisance	800	gpt-oss-120b	.95	.98	.02	.02	.90
BiasBios	nuisance	1589	qwen3-32b	.98	.98	.23	.02	.74
BiasBios	placebo	400	gpt-oss-120b	.96	.00	.01	.00	.98
BiasBios	placebo	961	qwen3-32b	.98	.00	.42	.00	.58
CEBaB	nuisance	210	gpt-oss-120b	.93	.98	.05	.22	.69
CEBaB	nuisance	210	qwen3-32b	.62	.97	.15	.32	.44
CEBaB	placebo	158	gpt-oss-120b	1.00	.00	.08	.00	.92
CEBaB	placebo	158	qwen3-32b	.91	.01	.51	.00	.46
CEBaB	target	125	gpt-oss-120b	.02	.10	.03	.16	.76
CEBaB	target	125	qwen3-32b	.00	.08	.10	.27	.55
Civil	nuisance	762	gpt-oss-120b	1.00	.97	.00	.62	.36
Civil	nuisance	762	qwen3-32b	.45	.98	.13	.72	.17
Civil	nuisance rev	400	gpt-oss-120b	1.00	.98	.00	.36	.62
Civil	nuisance rev	400	qwen3-32b	.63	.99	.09	.53	.34
Civil	placebo	480	gpt-oss-120b	1.00	.00	.06	.00	.94
Civil	placebo	480	qwen3-32b	.88	.00	.53	.00	.44
Civil	placebo group	400	gpt-oss-120b	1.00	.90	.00	.06	.94
Civil	placebo group	400	qwen3-32b	.80	.89	.06	.19	.64

Table 4: LLM-judge verdicts per edit kind.

data	edit kind	n	accept (Qwen3)	accept (gpt-oss)	both	κ
CEBaB	nuisance	210	.44	.69	.41	.41
CEBaB	placebo	158	.46	.92	.44	.07
CEBaB	target	125	.55	.76	.52	.42
BiasBios	nuisance	800	.76	.90	.72	.30
BiasBios	placebo	400	.60	.98	.60	.05
Civil	nuisance	762	.17	.36	.13	.31
Civil	nuisance rev	400	.34	.62	.29	.29
Civil	placebo	480	.44	.94	.43	.07
Civil	placebo group	400	.64	.94	.62	.08

Table 5: Agreement of the two judges' accept decisions on pairs both rated.

Robustness of CF$_N$. Table 6 gives rank correlations, across competitive extractors, between CF$_N$ computed on all vs. accepted edits; normalized by Δ_T vs. by the pooled within-cell score SD; signed vs. absolute; for the two edit directions separately (CEBaB, BiasBios); and for the forward vs. reverse Civil swaps. At $\rho_{\text{tr}} = 0.5$ on CEBaB the signed effect is small while the absolute change is not (median $|\text{CF}_N|$ .25 vs. signed .07, Qwen), which is why the direction-wise rankings agree less there. Placebo edits are smaller than CEBaB's human rewrites, so the placebo floor is not magnitude-matched for CEBaB.

data	model	ρ_{tr}	all vs. accepted	Δ_T vs. SD norm.	signed vs. $\lvert\cdot\rvert$	direction + vs. −	Civil fwd vs. rev
BiasBios	G	0.5	1.00	+.99	−.93	+.92	–
BiasBios	G	0.7	1.00	+.99	+.94	+.97	–
BiasBios	G	0.9	1.00	+.99	1.00	+.99	–
BiasBios	Q	0.5	+.99	+.99	−.54	+.94	–
BiasBios	Q	0.7	1.00	+.99	+.96	+.98	–
BiasBios	Q	0.9	1.00	+.97	1.00	+.99	–
CEBaB	G	0.5	+.77	1.00	+.13	+.31	–
CEBaB	G	0.7	+.93	+.99	+.89	+.78	–
CEBaB	G	0.9	+.99	+.97	+.99	+.94	–
CEBaB	Q	0.5	+.88	+.99	+.59	+.69	–
CEBaB	Q	0.7	+.98	+.99	+.97	+.94	–
CEBaB	Q	0.9	+.99	+.87	+.99	+.94	–
Civil	G	0.5	+.98	+.99	−.24	–	+.82
Civil	G	0.7	+.99	+.99	+.95	–	+.92
Civil	G	0.9	+.98	+.94	+.99	–	+.76
Civil	Q	0.5	+.93	+.97	+.42	–	+.76
Civil	Q	0.7	+.89	+.97	+.99	–	+.68
Civil	Q	0.9	+.92	+.93	1.00	–	+.68

Table 6: Rank correlations between variants of CF$_N$ across competitive extractors (medians of 3 training samples).

		best DEC	#near (equiv.)	CF$_N$ near-best range	CF$_N$ equiv. range	CONFLICT range
$\rho_{tr} = 0.7$						
CEBaB	Q	.934	18 (0)	.15–.54	–	.70–.86
	G	.967	13 (0)	.05–.26	–	.86–.93
BiasBios	Q	.995	62 (0)	.00–.13	–	.96–.99
	G	.995	61 (0)	−.01–.10	–	.96–.99
Civil	Q	.859	5 (0)	.14–.17	–	.64–.71
	G	.869	9 (0)	.09–.19	–	.66–.69

Table 7: As Table 2, at $\rho_{\text{tr}} = 0.7$ (equivalence tests were run at $\rho_{\text{tr}} \in \{0.5, 0.9\}$ only).

data	model	ρ_{tr}	CF$_N$	$\lvert\text{CF}_N\rvert$	CONFLICT	$\lvert$LEAK$\rvert$
BiasBios	G	0.5	−.32 [−.52, −.07]	+.10 [−.13, +.33]	+.90 [+.81, +.94]	−.24 [−.42, −.03]
BiasBios	G	0.7	+.41 [+.23, +.56]	+.40 [+.20, +.55]	+.69 [+.53, +.83]	+.59 [+.44, +.72]
BiasBios	G	0.9	+.39 [+.13, +.60]	+.39 [+.13, +.61]	+.52 [+.32, +.69]	+.40 [+.13, +.63]
BiasBios	Q	0.5	−.22 [−.45, +.05]	−.43 [−.62, −.22]	+.91 [+.84, +.95]	−.16 [−.38, +.10]
BiasBios	Q	0.7	−.01 [−.24, +.22]	−.01 [−.23, +.23]	+.75 [+.60, +.85]	+.35 [+.15, +.53]
BiasBios	Q	0.9	+.07 [−.24, +.34]	+.07 [−.22, +.35]	+.53 [+.32, +.70]	+.32 [+.04, +.53]
CEBaB	G	0.5	−.35 [−.57, −.03]	−.71 [−.83, −.51]	+.80 [+.64, +.89]	−.19 [−.46, +.10]
CEBaB	G	0.7	−.43 [−.67, −.15]	−.61 [−.77, −.36]	+.66 [+.45, +.81]	−.29 [−.55, +.02]
CEBaB	G	0.9	−.61 [−.75, −.42]	−.65 [−.78, −.46]	+.68 [+.52, +.78]	−.47 [−.63, −.27]
CEBaB	Q	0.5	−.23 [−.49, +.09]	−.65 [−.82, −.40]	+.74 [+.56, +.85]	−.08 [−.38, +.21]
CEBaB	Q	0.7	−.50 [−.67, −.23]	−.58 [−.74, −.36]	+.69 [+.51, +.79]	−.37 [−.57, −.11]
CEBaB	Q	0.9	−.43 [−.61, −.23]	−.46 [−.63, −.27]	+.35 [+.15, +.52]	+.20 [+.01, +.38]
Civil	G	0.5	+.05 [−.32, +.41]	−.51 [−.69, −.26]	+.67 [+.38, +.86]	+.43 [+.10, +.67]
Civil	G	0.7	−.20 [−.50, +.15]	−.30 [−.58, +.02]	+.24 [−.07, +.51]	+.53 [+.21, +.82]
Civil	G	0.9	+.03 [−.20, +.26]	−.02 [−.26, +.22]	−.25 [−.51, +.05]	+.66 [+.42, +.85]
Civil	Q	0.5	−.09 [−.40, +.26]	−.42 [−.63, −.13]	+.70 [+.43, +.84]	+.17 [−.22, +.50]
Civil	Q	0.7	−.01 [−.36, +.35]	−.08 [−.38, +.26]	+.24 [−.07, +.54]	+.53 [+.17, +.79]
Civil	Q	0.9	−.01 [−.29, +.29]	−.00 [−.30, +.33]	−.47 [−.69, −.20]	+.74 [+.49, +.91]

Table 8: Spearman correlation of test DEC with each facet across competitive standard extractors, with 95% bootstrap CIs over extractors. Hyperparameter variants of one family are not independent, so CIs are descriptive.

B Further results

Readout ablation. Reading the last non-special token instead of mean pooling (Qwen2.5-7B, Table 10) preserves the pattern: under correlated training, near-best extractors differ substantially in CF$_N$, decorrelated training largely removes it, and Conflict-based selection lowers CF$_N$. On Civil the last-token readout is less decodable and the selection costs considerably more accuracy.

C Synthetic representation

We generate $x = T\,u + N\,v + \varepsilon$ in $\mathbb{R}^{256}$ with unit vectors u, v at a chosen cosine ("overlap"), binary T and N with training agreement ρ_{tr}, and anisotropic Gaussian noise (power-law spectrum). A nuisance edit is exact: $x' = x \pm v$. We fit the same extractors with the same code and compute CF$_N$ on a decorrelated pool. With no overlap, CF$_N$ grows with training correlation (from ≈ 0 at $\rho_{\text{tr}} = 0.5$ to ≈ 0.75 at 0.9) for every standard extractor: the familiar shortcut. With decorrelated training, CF$_N$ grows with overlap (by .83–.87 across overlaps for standard extractors) while decodability changes relatively little (by at most .03; .09 for the erased probe; Figure 3). In this

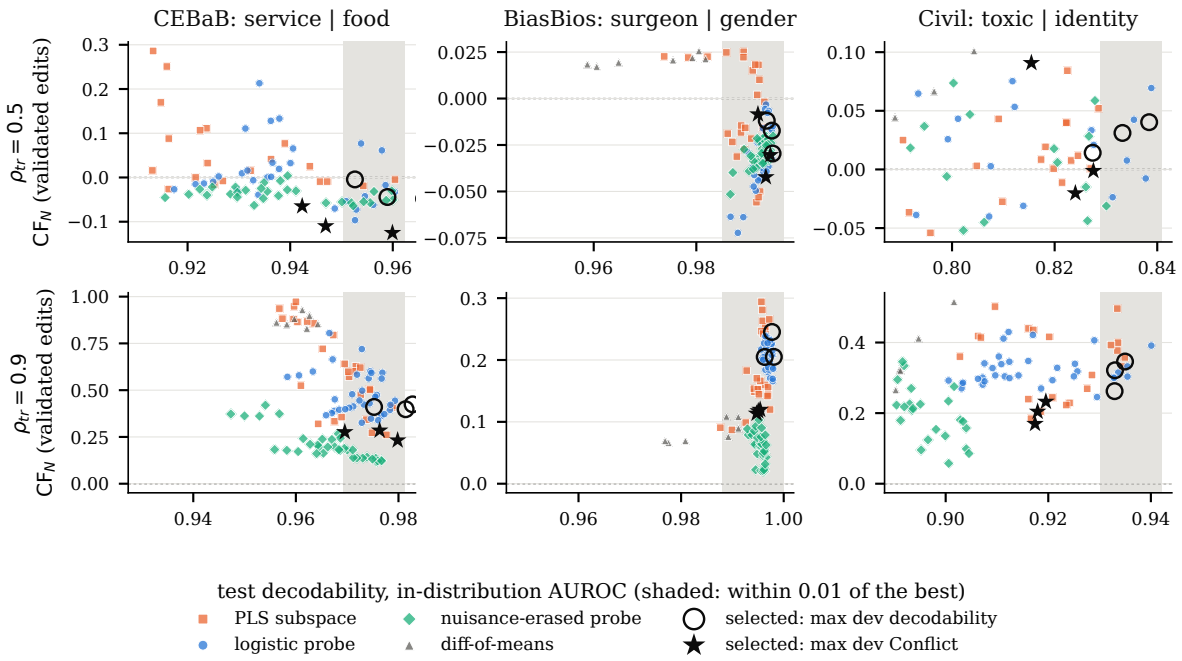


Figure 2: As Figure 1, for Gemma-2-9B-it.

data	model	ρ_{tr}	ΔDec [90% CI]	equiv.	CF$_N$ probe	CF$_N$ erased
BiasBios	G	0.5	−.000 [−.000, +.000]	yes	−.01	−.03
BiasBios	G	0.9	+.001 [−.001, +.002]	yes	.20	.06
BiasBios	Q	0.5	+.000 [−.000, +.000]	yes	−.00	−.02
BiasBios	Q	0.9	+.001 [+.000, +.002]	yes	.20	.06
CEBaB	G	0.5	+.000 [−.000, +.001]	yes	−.00	−.01
CEBaB	G	0.9	+.004 [−.003, +.012]	no	.63	.14
CEBaB	Q	0.5	−.003 [−.006, +.000]	yes	.13	.02
CEBaB	Q	0.9	+.009 [−.003, +.023]	no	.83	.34
Civil	G	0.5	+.008 [+.005, +.011]	no	.01	−.01
Civil	G	0.9	+.039 [+.031, +.049]	no	.26	.18
Civil	Q	0.5	+.011 [+.006, +.015]	no	.06	.04
Civil	Q	0.9	+.049 [+.036, +.062]	no	.40	.28

Table 9: Logistic probe vs. nuisance-erased probe, each family's layer and hyperparameter chosen on development Dec (training sample 0). Equiv.: 90% CI of the Dec difference within ±0.01.

		$\rho_{tr} = 0.9$		$\rho_{tr} = 0.5$		selection, $\rho_{tr} = 0.9$
data	readout	best Dec	CF$_N$ near-best	best Dec	CF$_N$ near-best	CF$_N$: Dec→Conflict (ΔDec)
CEBaB	mean	.966	.46–.89	.924	.02–.28	.69→.54 (−.012)
CEBaB	last token	.968	.51–.76	.921	.15–.27	.56→.49 (−.003)
BiasBios	mean	.998	.08–.41	.995	−.04–.04	.20→.12 (−.001)
BiasBios	last token	.998	.09–.66	.994	−.05–.01	.27→.09 (−.005)
Civil	mean	.932	.24–.45	.829	.04–.08	.43→.33 (−.018)
Civil	last token	.904	.67–.67	.794	.01–.05	.67→.09 (−.156)

Table 10: Readout ablation (Qwen2.5-7B, standard extractors, accepted edits): mean pooling vs. the last non-special token.

low signal-to-noise setting even logistic probes inherit the overlap; in CEBaB, where the target is decoded far better, they inherit much less of it (§4). The synthetic setting illustrates that decodability is largely insensitive to overlap; it is not a model of the real representations.

D Reproducibility and implementation audit

Code, data construction, judge outputs and per-extractor metrics are in the repository linked on the first page; every table and in-text number in this paper is generated from saved results by a script. An automated audit (released) checks that: training, development and test texts are disjoint (CEBaB at the level of review families; two BiasBios test texts and one development text that duplicated training texts were removed); every edit's orientation matches its labels; PLS with $k{=}1$ reproduces the difference of means numerically (cosine 1.000, same orientation); no evaluated text is truncated at 256 tokens on CEBaB and BiasBios, and 0.2% on Civil; and an independent re-implementation of every metric from the saved raw scores (rank-based AUROC, no shared code) matches the pipeline to $< 10^{-8}$. Layers: Qwen hidden states 7, 10, 14, 18, 21, 24, 28 of 28; Gemma 10, 16, 21, 26, 32, 37, 42 of 42. Bootstrap CIs resample review families (CEBaB) or texts within (T, N) cells, and edit pairs.

We do not report residual "completeness" (Elazar et al., 2021): whether linear information survives projecting out a subspace is largely fixed by whether the subspace contains the class-mean difference (Belrose et al., 2023), a property of the estimator rather than of what its score measures.

		standard extractors: CF_N (ΔDEC)				+ erased probe: CF_N (ΔDEC)		
		max DEC	max CONFLICT	min |LEAK|	DEC−|LEAK|	max CONFLICT	min |LEAK|	DEC−|LEAK|
CEBaB	Q	.69	.54 (−.012)	.50 (−.013)	.52 (−.008)	.34 (−.007)	.34 (−.007)	.33 (−.011)
	G	.41	.26 (−.005)	.27 (−.004)	.26 (−.005)	.11 (−.004)	.12 (−.004)	.14 (−.010)
BiasBios	Q	.20	.12 (−.001)	.08 (−.004)	.10 (−.021)	.03 (−.002)	.03 (−.002)	.03 (−.002)
	G	.22	.12 (−.002)	.11 (−.006)	.07 (−.017)	.02 (−.002)	.02 (−.001)	.02 (−.001)
Civil	Q	.43	.33 (−.018)	.31 (−.001)	.37 (−.041)	.15 (−.034)	.31 (−.001)	.17 (−.038)
	G	.31	.20 (−.015)	.30† (−.006)	.22† (−.020)	.11 (−.036)	.30† (−.006)	.11 (−.034)

Table 11: All development-set selection criteria at $\rho_{\text{tr}} = 0.9$ (as Table 3), for standard extractors and with the nuisance-erased probe.

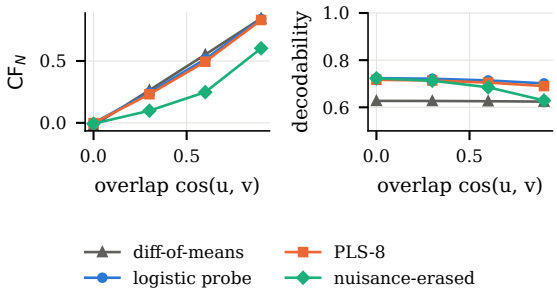


Figure 3: Synthetic representation, decorrelated training ($\rho_{\text{tr}} = 0.5$): nuisance sensitivity (left) and decodability (right) as target–nuisance overlap grows; means over 5 replicates.